

AGENTPROF: Semantic Profiling for AI Agents

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Abstract

AI agents orchestrate long-running activities across users, tools, and system resources, producing growing populations of execution histories. Developers need population-level answers about where failures concentrate, which workflows trigger unsafe effects, and which task categories consume the most budget. Unlike stable code paths, natural-language intent lacks shared identity, and native execution nesting does not provide a reusable cross-run hierarchy of semantic responsibility. Existing agent tools trace runs, group metadata, and aggregate metrics, but do not recursively construct shared semantic responsibility while retaining LLM/tool evidence and conserved resource measures in a standard profile. Agent observability needs profiling, not only debugging. AGENTPROF realizes this insight through a *semantic operation stack model*: backend-neutral recursive annotations identify shared responsibilities over an ordered source tree, operation stacks compose those responsibilities with native evidence, and additive resource views emit pprof-compatible profiles. We evaluate AGENTPROF on 405 human-staged CodeTraceBench trajectories, three complete public localization benchmarks, 287 OSWorld-Human sessions, two population case studies, and four public cost workloads. Automatic Agent annotation raises ordinary B³ F1 against human stages from 0.541 for raw action and 0.663 for recurrence to 0.704, while semantic grouping improves MAP over raw action in all three localization benchmarks. Evaluated task-family and action backends reach 0.695 and 0.498 macro-F1, label-free recurrence reaches 0.680 boundary F1 and 0.786 B³ F1 on OSWorld-Human, and AGENTPROF constructs a 27,765-operation profile in 1.17 s.

Introduction

AI agents (Yang et al. 2024; OpenAI 2025; Anthropic 2025; Xie et al. 2024) orchestrate multi-step activities across a high-level intent layer (prompts, LLM calls, and tool invocations) and a low-level system-effects layer (process spawns and file and network I/O). As agents evolve from short scripts into complex systems, production teams accumulate many trajectories.

Developers lack population-level answers about agent quality, safety, and cost. Which categories of tasks consume

the most token budget? Where do failures concentrate across workflows? Which behavioral patterns trigger unsafe system effects? At scale, manual trajectory inspection is slow and expensive (Lù et al. 2025), while LLM judging requires a separate evaluator pass per trajectory (Lù et al. 2025; Zheng et al. 2023).

Agent trajectories lack two structures needed for cross-run profiling. Responsibility lies in semantic categories such as task intent and tool operation rather than stable function names. Natural-language prompts provide no common identifier for shared intent. Native execution nesting may exist, but it does not provide a reusable cross-run hierarchy of semantic responsibility. A prompt sequence may represent multiple tasks, while one task may belong to a larger task. Thus adapting profiling requires stable semantic identifiers and an attribution hierarchy that does not depend on a run-time stack.

Existing agent observability lacks this profiling abstraction. Tools provide rich tracing, metadata aggregation, and population dashboards (LangChain 2024; Langfuse 2024; OpenTelemetry 2024; Datadog 2025), and some derive hierarchical categories across traces and aggregate latency, cost, or evaluation metrics (LangChain 2026; Datadog 2026), while recent research canonicalizes actions into process profiles or cross-run graphs for monitoring and recovery (Shu et al. 2026; Liu et al. 2026a; Zhong, Saxena, and Raghunathan 2026; Nian et al. 2026). These systems establish semantic grouping, metric rollups, and cross-run analysis. They do not combine recursively annotated semantic responsibility, source-linked LLM/tool evidence, conserved additive measures, and standard profiler output over the same histories.

Agent observability needs profiling, not only debugging. Profiling applies to agents when a profiler projects resource measures onto stable semantic tags in an attribution hierarchy. Our *semantic operation stack model* provides this structure through source nodes, recursive operation annotations, and operation stacks. Source nodes retain native session, prompt, LLM-call, and tool-call relations and additive measures. An interchangeable backend names the session scope, each user-prompt scope, and recursively refined intervals as operations. An *operation stack* emits the agent, those semantic operations, and the covered LLM/tool evidence; raw session and prompt IDs remain labels rather than visible

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frames that fragment aggregation (Figure 2).

We implement this model in **AGENTPROF**, an offline profiler that compiles local agent trajectories into pprof-compatible profiles (Google 2024) (Figure 1). *Operation annotation* lets an Agent, a language model, or a non-LLM backend mark nested source intervals with shared semantic names. *Stack construction* deterministically emits `agent`→`session-level operation`→`prompt-level operation`→`recursive operations`→`LLM call`→`tool call`. Raw session and prompt identities remain pprof labels for drill-down.

Across all 405 reconstructable failed CodeTraceBench trajectories (Li et al. 2026), automatic Agent annotation reaches 0.704 ordinary B³ F1 against human stages, compared with 0.663 for multi-resolution recurrence and 0.541 for raw action. Across three complete public workloads, semantic grouping raises MAP over raw action from 0.773 to 0.789, 0.281 to 0.452, and 0.121 to 0.230, respectively (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). On 287 OSWorld-Human development sessions (Abhyankar, Qi, and Zhang 2025), label-free recurrence reaches 0.680 boundary F1 and 0.786 B³ F1. On the AgentBoard and ASE populations, task-family and action backends reach 0.695 and 0.498 macro-F1, respectively (Ma et al. 2024; Bouzenia and Pradel 2025). On four public workloads, **AGENTPROF** constructs a 27,765-operation profile in 1.17 s.

This paper makes three contributions:

1. **Semantic operation stack model.** Recursive semantic responsibility over source-native evidence supplies agent observability’s missing profiling layer (Background and Motivation; Design).
2. **System: AGENTPROF.** A backend-neutral annotation workspace composes recursively marked semantic intervals with source-native hierarchy and produces pprof-compatible profiles (Implementation).
3. **Evaluation.** Automatic operation structure improves human-stage agreement and problem ranking over raw action. Two population case studies expose high-cost unfinished work and success–failure differences, and **AGENTPROF** profiles 27,765 operations in 1.17 seconds (Evaluation).

Background and Motivation

LLMs and AI Agents

A typical agent trajectory interleaves two layers of activity (Yang et al. 2024; OpenAI 2025; Anthropic 2025; Zheng et al. 2025). At the intent layer, an agent receives prompts, issues LLM calls, and invokes tools for code execution, search, and file editing. Tool invocations trigger lower-layer process, file, and network effects (Zheng et al. 2025). Trajectories repeat this cycle, and teams accumulate many trajectories.

System Profiling

Unlike single-execution debugging, profiling aggregates measurements to attribute resources to responsible entities. Traditional profilers sample execution, attach each sample to its call stack, and fold identical stacks (Google 2024;

Gregg 2011). Flame graphs (Gregg 2011) and pprof (Google 2024) call graphs visualize aggregate cost by width or node size, while Pivot Tracing (Mace, Roelke, and Fonseca 2015) dynamically selects and groups measurements across causally related events. These tools start from stable code identifiers and runtime stacks, although pprof can also promote `tagroot`/`tagleaf` values to pseudo-frames (Google 2024). Agent profiles must instead derive stable semantic fields and connect them to downstream effects.

Challenges for Agent Profiling

Existing tools (LangChain 2024; Langfuse 2024; OpenTelemetry 2024) record agent events as per-execution span trees for single-run debugging. AgentSight (Zheng et al. 2025) bridges intent and system-effect layers by correlating extended Berkeley Packet Filter (eBPF) system events with intercepted LLM traffic, but the resulting recordings still lack aggregate profiling.

Agent profiling faces two challenges. First, responsibility lies in semantic categories such as task intent and workflow phase, not code paths. Standards represent application-supplied span attributes (OpenTelemetry 2024; Arize AI 2024), but natural-language histories lack stable categories because different prompts or tools can express the same intent.

Second, agent trajectories lack a runtime call hierarchy for semantic attribution. In CPU profiling, the stack attributes `malloc`’s cost simultaneously to `parse`, `process`, and `main` (Google 2024; Gregg 2011). A prompt sequence may contain one or several tasks within a larger workflow, yet no runtime mechanism marks its boundaries or attribution granularity. RQ1 evaluates how this missing hierarchy affects attribution, while RQ3 evaluates constructed groups and boundaries.

Design Requirements for Agent Profiling

Traditional profiling relies on three mechanisms that the call stack supplies automatically. Agent profiling requires all three without a call stack:

- D1: Conserved multi-resource attribution.** Each atomic source record must retain additive resource measures, such as tokens, time, file effects, and network effects, so regrouping never changes their total mass.
- D2: Shared semantic identity.** Responsibilities that recur in different sessions need the same short semantic name so that their costs fold together rather than remaining session-local.
- D3: Hierarchy without evidence loss.** The profiler must derive variable-depth semantic responsibility from the native trace while retaining covered LLM/tool evidence as leaf frames and raw session/prompt identities as drill-down labels.

Design

The design has three explicit objects: an ordered source tree, nested semantic annotations, and a weighted pprof stack. A source adapter constructs the first without discarding content

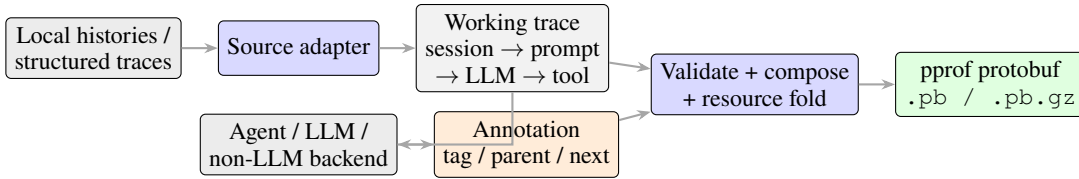


Figure 1: AGENTPROF pipeline. A source adapter preserves the native trace tree. An Agent, LLM, or non-LLM backend writes the same small annotation contract. The CLI validates and composes semantic paths with source evidence, folds one additive resource measure, and emits only a standard pprof profile.

or native parent relations. Any annotation backend constructs the second without rewriting source nodes. The deterministic profiler validates both, emits agent and semantic-operation frames followed by covered LLM/tool evidence, and folds equal paths into the third.

This separation addresses D1–D3 directly. Additive measures remain attached to source nodes; shared semantic names create cross-session identity; LLM/tool frames preserve visible evidence; and unique session, prompt, call, and event IDs remain pprof labels. Switching from operation count to tokens, time, file effects, or network effects replays the same annotation and changes only stack width.

Semantic Operation Stack Model

Let a source node be $n = (id, parent, kind, data, m)$ in an ordered forest. The *kind* is a session, prompt, LLM call, tool call, effect, or an adapter-defined atomic event. The *data* retains the source content needed to interpret the event, and *m* maps resource names to nonnegative additive measures. Source-parent links encode observed execution structure; they never encode an inferred semantic operation.

Recursive annotation covers every source node with exactly one active semantic leaf and its enclosing operation ancestors; let this path be $A(n)$. Let $g(n)$ be the agent identity inherited from the source root, and let $E(n)$ be the retained evidence ancestry after removing raw session and prompt containers—normally the covered LLM call, tool call, and effect. The emitted operation stack is $\sigma(n) = \langle agent : g(n) \rangle \parallel A(n) \parallel E(n)$, where $\parallel$ denotes concatenation. Raw session and prompt IDs are attached as labels $\lambda(n)$ rather than visible frames. Equal semantic prefixes can therefore aggregate across sessions without losing the evidence needed to locate a contributing call or tool event.

We formalize a resource profile as (φ, σ, w_r) : predicate $\varphi(n)$ selects source nodes, $\sigma(n)$ supplies the composed stack, and $w_r(n) = m_n[r] \geq 0$ selects one additive resource. Folding merges equal stacks and sums w_r . The default operation-count view assigns one unit to each adapter-defined atomic operation; token, time, file-effect, and network-effect views select their corresponding measures. Because A and E remain fixed, changing r changes only widths, not boundaries or semantic names.

Recursive Operation Annotation

The source adapter preserves an ordered tree of sessions, prompts, LLM calls, and tool calls. Semantic operations are nested intervals over this tree rather than replacements for its

nodes. An annotation at source node s contains a semantic name, the start node of its enclosing annotation, and the first source node after its interval. The exclusive end may be omitted when the interval reaches its parent’s end. Every source root must begin a session-level operation, and every prompt must begin a prompt-level operation. Further nested intervals define the variable-depth suffix of the operation path. Equal path prefixes fold across sessions; covered LLM/tool nodes remain below the path, while raw session and prompt IDs remain labels for evidence drilldown.

Annotation proceeds coarse to fine. A backend first names the complete session and each user-prompt scope, then selects one interval whose current name hides a useful responsibility change, marks source-visible boundaries, names the resulting child intervals, and repeats only where another distinction helps. Branches stop independently, so depth is neither fixed nor optimized directly. A later iteration may rename, merge, or remove an earlier split. The default Agent-assisted backend reads source content and existing paths to make these decisions. A plain language model, deterministic rules, change-point methods, and the label-free recurrence constructor use the same annotation contract. The profiler therefore does not depend on one model or inference service.

Implementation

AGENTPROF is an offline Rust command-line interface that implements the semantic operation stack model (Figure 1).

Input reconstruction. Source-specific adapters reconstruct an ordered tree from local Codex and Claude Code histories, AgentSight (Zheng et al. 2025) recordings, or public trajectory datasets. Each adapter preserves replay-stable node IDs, source content or a readable reference to it, native parent links, input order, and available additive measures. Structured operation JSONL is accepted by materializing its records as adapter-defined atomic source nodes rather than inventing missing prompt or call relations.

Annotation workspace. The backend reads `trace.jsonl` and writes only `annotation.json`. The former retains source content, source-parent links, additive measures, and the current derived path; the latter maps source start IDs to `tag`, semantic `parent`, and exclusive `next`. AGENTPROF validates that references exist and intervals are nested, noncrossing, and collectively cover every source node, then atomically updates derived paths and the current folded aggregate. It also emits nonblocking hierarchy warnings for a recursive operation with only one

explicit semantic child and for a large flat fan-out with little recursive refinement. These diagnostics expose redundant or prematurely flat annotations without forcing artificial depth or blocking profile generation. The same annotation can be replayed with operation count, tokens, time, file effects, or network effects without changing a boundary or semantic name.

Non-LLM recurrence backend. The Rust inducer counts adjacent action transitions in reference sessions and scores each transition by normalized pointwise mutual information (NPMI), $\text{NPMI}(a, b) = \ln[p(a, b)/(p_L(a)p_R(b))]/-\ln p(a, b)$, with all three probabilities defined over the same transition sample space (Bouma 2009).

Occurrence-weighted one-dimensional k -means with $k = 2$ (MacQueen 1967) initializes at the minimum and maximum scores, assigns distance ties to the lower center, and uses the converged midpoint as a global cutoff. The inducer applies the same procedure to action-changing transitions for a second cutoff. Same-action pairs use the global cutoff, whereas different-action pairs use the smaller cutoff, which can remove a boundary produced by the global rule but cannot add one. When every reference and target operation also has a nonempty `action_detail`, the inducer fits the same recurrence model to the compound signature. Detailed continuity may remove a coarse boundary but can never add one. Missing, unseen, or weak detail falls back exactly to the coarse decision. Segment names remain coarse run-length-compressed action sequences. The inducer reads only session identity, input order, action, and optional action detail, not target annotations or resource weights.

An optional reference-calibrated recurrence mode uses the same NPMI score but selects one scalar cutoff that maximizes per-operation B^3 F1 on disjoint reference groups. The mode enumerates score intervals, resolves exact ties toward the smallest cutoff, and never reads target groups. Label-free k -means remains the default.

Profile export. AGENTPROF emits one pprof protobuf profile for existing tools such as `go tool pprof`; it does not add a visualization frontend or a second export format.

Evaluation

Research questions. The evaluation addresses four research questions: **RQ1:** Does one semantic hierarchy expose different resource bottlenecks? **RQ2:** Does profiler output correspond to real problems? **RQ3:** How accurately do automatic backends recover operation structure? **RQ4:** What is the cost of constructing a semantic profile? Each RQ subsection pairs its protocol with the strongest answer its evidence supports.

We evaluate three data classes separately. **Real inputs (RQ1 and case studies)** include 41 long-horizon coding-agent sessions and 440 real mixed-outcome web-agent trajectories. **Annotated trajectories (RQ3 and RQ4)** include 15 real-agent or human web, API, coding, and mobile/GUI execution families, plus 405 CodeTraceBench trajectories with 20,866 operations (Lù, Kasner, and Reddy 2024; Deng et al. 2023; Xu et al. 2025; Yao et al. 2022; Li et al. 2023;

Qin et al. 2024; Yao et al. 2025; Yang et al. 2024; Li et al. 2024; Lu et al. 2025; Liu et al. 2026b; Lù et al. 2025; Chen et al. 2026b; Wang et al. 2025; Abhyankar, Qi, and Zhang 2025). **Problem-localization benchmarks (RQ2)** provide independent targets over 27,346 operations (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). The RQ4 union contains 27,765 operations (Lù et al. 2025; Chen et al. 2026b; Abhyankar, Qi, and Zhang 2025; Wang et al. 2025).

RQ1: Multi-Resource Attribution

RQ1 asks whether a fixed semantic hierarchy reveals materially different bottlenecks when the additive resource measure changes. The source population contains 41 real long-horizon agent trajectories, 3,146 source-native turns, and 5,750 operations. Within it, three independent executions repeat the same Git-deployment task. The automatic Agent backend assigns 96 recursive annotations to their 735 source nodes. The resulting hierarchy reaches semantic depth five without a fixed limit, and the CLI reports no unary or flat-fan-out warning.

Case Study 1: Long-Horizon Agent Diagnosis. Figure 2 uses a stock-pprof focus query to retain all three executions of one repeated Git-deployment task rather than selecting one trace.

The focused task contains 489 operations and 4,558,192 provider-reported tokens from OpenHands/Claude, OpenHands/DeepSeek, and Terminus2/DeepSeek. By count, Terminus2 contributes 56.24% and OpenHands 43.76%; by tokens, OpenHands contributes 86.62%. The repeated `diagnose rejected SSH password authentication` responsibility accounts for 97 operations but 1,936,828 tokens (19.84% and 42.49% of the focused task, respectively). Source drilldown shows both OpenHands executions returning to this responsibility after control or fallback attempts; all three executions verify web deployment or substitute transport paths, but none establishes the requested password-authenticated `git@localhost` path. Thus, the fixed hierarchy exposes a bottleneck that an operation-count view substantially understates: the SSH-diagnosis path is roughly one fifth of work by count but more than two fifths by tokens. Source drilldown explains the disparity and shows the expensive path did not establish the requested terminal condition. RQ1 therefore answers yes for this repeated real task without claiming that one resource measure universally dominates another.

RQ2: Problem Correspondence

RQ2 asks whether target-blind profiles place independently annotated problems in groups a developer can inspect early. We run complete AgentProcessBench, HINTBench, and TraceElephant workloads (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). Of HINTBench’s reported 629 trajectories, the released test snapshot used here contains 536. A separate 80-trajectory validation snapshot selects one of 24 field orders before all 536 tests are scored.

Within each workload, we give AGENTPROF and raw-action grouping identical operations and precomputed benchmark



Figure 2: All three real `git-multibranch` executions under one fixed operation hierarchy. Width is operation count (top) or provider-reported tokens (middle); the bottom panel focuses the repeated SSH-password diagnosis and exposes its recursively nested hypotheses and controls. The profiles retain LLM/tool evidence frames and source labels; the paper rendering hides only the leaf evidence frames to keep the operation hierarchy legible. AgentPProf emits standard pprof, and these static panels use the official Brendan Gregg FlameGraph tool rather than a custom frontend.

Workload	Sem.	Raw	Agent+Evidence	Agent-only
AgentProcessBench	.789	.773	.773	.730
HINTBench	.452	.281	.414	.284
TraceElephant	.230	.121	.252	.194

Table 1: Standard per-query AP averaged as MAP after scoring every operation. Sem. is the declared semantic hierarchy. Agent+Evidence composes identical automatic Agent marks with retained LLM/tool/action evidence frames; Agent-only omits those frames. Higher is better.

judge/localizer predictions, so they differ only in grouping. We fix predictions, group scores, profiles, and scoring rules before loading the corresponding test target labels. We select HINTBench’s field order on the separate validation snapshot described above. AgentProcessBench averages judge votes within each group. For HINTBench and TraceElephant, we assign every root-to-frame stack prefix the 95% Wilson lower bound (Wilson 1927) of the fraction of its members with a positive frozen benchmark prediction. Each operation inherits the maximum score among prefixes containing it, with ties preserved.

Each query is one trajectory containing an annotated target. We report non-interpolated per-query AP, evaluating ties at their shared score threshold, and arithmetic-mean MAP across the 614, 400, and 220 target-bearing queries, respectively (Robertson 2008).

In the primary comparison, AGENTPROF improves MAP over raw action on all three complete workloads (Table 1). The gains are +.016 [.005,.027], +.171 [.154,.188], and +.109

[.077,.142], using 10,000 paired resamples of trajectory clusters within benchmark-defined strata.

The automatic Agent marks cover all 1,000/536/220 sessions and 8,509/12,877/5,960 operations. Retaining source-call frames beneath those same marks raises MAP over Agent-only by .043, .130, and .058, respectively; the paired task-cluster intervals exclude zero in all three workloads. The result directly supports the design decision that semantic operations retain source evidence rather than discarding it. Agent+Evidence approaches the declared semantic hierarchy on the first two workloads and exceeds it on TraceElephant.

Case Study 2: Differential Profiling at Scale

We next aggregate the complete mixed-outcome population from AgentRewardBench (Lù et al. 2025), rather than presenting one favorable trace pair. The collection contains 440 real trajectories over 125 tasks with both successful and unsuccessful runs, producing 338 complete bad-good pairs: 24 pair occurrences from AssistantBench, 102 from VisualWebArena, 144 from WebArena, and 68 from WorkArena. Because tasks can contain multiple runs, the 202 successful and 238 unsuccessful sessions are reused across pairs; the profile is therefore pair-occurrence weighted. All 338 bad members and all 338 good members are folded into one signed operation-count pprof. Individual pair profiles are used only to descend from a collection-level path to source evidence.

The aggregate contains 7,366 bad-side and 3,780 good-side operation occurrences. Positive values denote bad-only or bad-side excess paths; negative values denote good-only or good-side excess paths. The largest bad-side result paths are continuing progress (+1,825) and exact re-

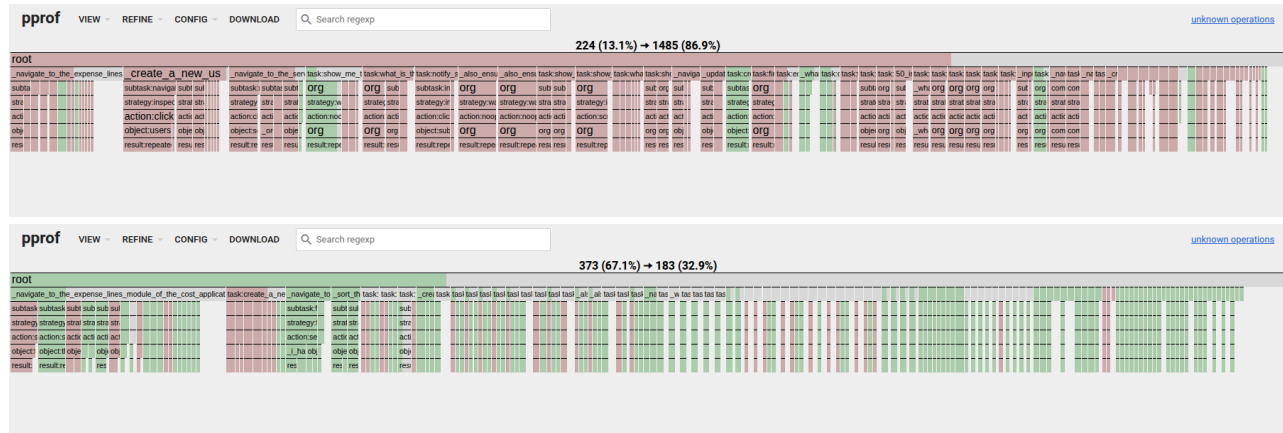


Figure 3: Focused stock-pprof views of the signed differential profile over all 338 bad-good pairs. Red is bad-side excess and green is good-side excess. The repeated-work path contains 1,485 bad-side versus 224 good-side occurrences (top); completion paths contain 183 bad-side versus 373 good-side occurrences (bottom). These are aggregate diagnostic differences, not causal effects.

peated action-state work (+1,261). The latter decomposes into click (+639), no-op (+356), and scroll (+277) paths. Concrete bad-side error families remain separately searchable: clicking an invisible element (+148), click timeout (+129), stopped execution (+92), filling a non-input element (+57), and selecting on a non-select element (+46). The good side contains more terminal (-92) and conclusion (-67) paths; `send_msg_to_user` has a cumulative difference of -100.

This is diagnostic structure, not a new outcome classifier. On the same population, simple step count remains the strongest tested scalar (pairwise accuracy .7633; trajectory ROC AUC .7517). The profile adds a different answer: it identifies whether excess work is repetition, waiting, wrong-widget interaction, timeout, stopping, or missing completion, while retaining action and object frames for stock pprof drill-down. AgentRewardBench has outcome and looping annotations but no gold semantic hierarchy, so this case establishes complete differential-profile coverage and concrete path exposure, not semantic-stack accuracy or causality.

RQ3: Automatic Operation Structure

RQ3 evaluates automatic backend outputs against independent annotations under each protocol’s stated input boundary. Literal task and action fields use accuracy and macro-F1 (the unweighted mean of per-class F1) (Lewis et al. 2004). Partition-valued task, phase, and group fields use V-measure (Rosenberg and Hirschberg 2007) or ordinary B^3 , and adjacent group boundaries use exact precision, recall, and F1 (Ruokolainen et al. 2016). Literal labels, permutation-invariant partitions, and adjacent boundaries are distinct outputs. A backend output is evaluated at the level it predicts: literal names, permutation-invariant leaf partitions, or adjacent interval boundaries. For legacy field-valued datasets, a mechanical adapter converts each predicted group into the same interval-annotation contract before AGENTPROF folds the original additive weights.

We first evaluate the default Agent-assisted backend on all

Method	B^3 P	B^3 R	B^3 F1	Bound. F1
Automatic Agent (A2)	0.839	0.607	0.704	0.394
Multi-resolution recurrence	0.782	0.575	0.663	0.266
Native source tree	0.975	0.249	0.397	0.259
Source-native turn	0.983	0.221	0.361	0.246
Raw-action grouping	0.891	0.388	0.541	—

Table 2: Automatic operation-structure agreement on all 405 CodeTraceBench trajectories. B^3 scores the leaf partition; boundary F1 scores exact adjacent transitions.

405 reconstructable CodeTraceBench trajectories (Li et al. 2026): 20,866 operations and 2,948 human-verified contiguous stages across four agent frameworks. Automatic Agents receive source-only packets and emit sparse complete-path marks over 17,148 source-native turns. A deterministic representation repair attaches a one-turn task-root-only prefix to its first nested responsibility, eliminating 149 artificial leaf groups without reading a stage label. The final 5,752 marks have semantic depth one (51 marks), two (5,608), or three (93); no prompt requests a depth.

Ordinary operation-level B^3 measures the induced leaf partition, while exact adjacent-boundary F1 measures transition placement. Both are standard metrics and assign every operation equal weight. The recurrence and source-native controls receive the same operation sequence.

Automatic Agent annotation reaches 0.704 B^3 F1, improving over recurrence by 0.0414 (task-clustered bootstrap 95% interval [0.0214, 0.0606]) and over raw action by 0.163. Its boundary F1 is 0.394 versus 0.266 for recurrence. Thus the Agent-assisted backend is the strongest tested automatic constructor on this complete population, and its marks preserve exactly 20,866 operation counts and 494,862,929 provider-reported tokens when replayed at either width.

We test all 287 OSWorld-Human task-instance sessions (Abhyankar, Qi, and Zhang 2025) with complete

Method	Prec.	Recall	Bound. F1	B ³ F1
Supervised predictor	0.700	0.782	0.739	0.816
Reference-calibrated recurrence	0.610	0.922	0.734	0.801
Label-free recurrence	0.592	0.799	0.680	0.786
Always boundary	0.476	1.000	0.645	0.678
Action change	0.386	0.626	0.477	0.659
Phase change	0.441	0.268	0.334	0.665

Table 3: RQ3 boundary and partition agreement on OSWorld-Human. Boundary F1 scores exact adjacent-pair decisions. B³ scores predicted-human partitions.

group annotations. This population contains 3,978 operations, 3,691 adjacent pairs, and 2,042 groups. For the supervised Bernoulli Naive Bayes predictor (McCallum and Nigam 1998), we fix the model family, nine input fields, adjacent-pair features, and training procedure. Each session-held-out cross-validation fold fits its parameters and threshold on training sessions and predicts every held-out session once. Predicted boundaries become an ordinary group field that AGENTPROF projects and folds. With the same five folds, label-free recurrence uses only the other sessions’ visible action transitions. OSWorld-Human supplies no non-redundant action detail, so the multi-resolution constructor exactly falls back to its coarse arm. Because we designed the recurrence rule after inspecting earlier results on this corpus, these results are development evidence rather than independent cross-family confirmation. Optional reference-calibrated recurrence uses training operations and group annotations to fit one NPMI cutoff, while held-out folds contribute only visible actions until predictions are fixed.

Controls place a boundary after every operation, visible action change, or visible phase change. We also report ordinary per-operation B³ partition F1 (Bagga and Baldwin 1998), whose predicted-human group overlap captures merging and fragmentation.

The supervised predictor reaches 0.739 boundary F1 and 0.816 B³ F1 versus 0.645 and 0.678 for the strongest controls. Label-free recurrence reaches 0.680 and 0.786, while reference-calibrated recurrence reaches 0.734 and 0.801 but requires group annotations that the default label-free method does not use. Every method conserves the total projected weight.

For target-blind TF-IDF/K-Means, V-measure (Rosenberg and Hirschberg 2007) over operation assignments reaches 0.557 on nine Mind2Web sessions (49 operations) and 0.815 on 100 ScienceWorld sessions (2,504 operations) (Deng et al. 2023; Wang et al. 2022). Every operation receives a prediction in both datasets, whereas a constant tag yields V-measure 0. Here, V-measure tests partitions, not literal names.

For task-family tags, we use a fixed evaluation-only Qwen3.6-27B llama.cpp backend distinct from the 3B CLI backend. Using only goals and declared family descriptions, it classifies all 1,012 AgentBoard goals into nine families (Qwen Team 2026; Ma et al. 2024). The backend reaches 0.695 macro-F1 and 0.733 accuracy, compared with 0.044 macro-F1 and 0.248 accuracy for the majority-class baseline, and produces identical assignments across three runs.

For action tags, a standalone backend-level adapter runs the fixed Qwen3.6-27B closed-label configuration using only the current thought/action and eight action definitions operationalized by TraceView (Qwen Team 2026; Sajadi et al. 2026). The adapter classifies all 2,737 published annotations from 120 AutoCodeRover, OpenHands, and RepairAgent trajectories (Bouzenia and Pradel 2025). The model reaches 0.498 macro-F1 and 0.628 accuracy, compared with 0.061 macro-F1 and 0.323 accuracy for the majority-class baseline. Its 0.437 macro-F1 gain has a whole-trajectory 95% bootstrap interval of [0.380, 0.494], and both complete runs agree exactly. Thirty-nine AutoCodeRover inputs contain the target word `Locate`. Excluding them leaves 0.490 macro-F1 and 0.622 accuracy, so the positive comparison to majority is unchanged.

Thus, across the named public populations, the automatic Agent constructor reaches 0.704 B³ and 0.394 boundary F1 on CodeTraceBench; the label-free recurrence backend reaches 0.786 B³ and 0.680 boundary F1 on OSWorld-Human; and closed-label task-family and action backends reach 0.695 and 0.498 macro-F1. These protocols evaluate complementary structural and literal outputs rather than collapsing them into one bespoke score.

RQ4: Profiling Cost

RQ4 measures the profiling core: parsing fixed operation JSONL, constructing stacks, folding weights, and serializing the profile. It excludes capture, source adaptation, and automatic annotation. Across four complete public workloads and their union, we compare the fixed semantic hierarchy project, agent, task, phase, op, tool, status with the raw hierarchy project, agent, action, status on identical inputs over three runs. Measurements use agentpprof 0.2.37 on a 24-core Intel Core Ultra 9 285K with 125 GiB RAM and Linux 6.15.11.

Workload	Ops	Semantic (s)	Raw action (s)	Peak RSS (MiB)
AgentRewardBench	729	0.04	0.03	19.0
SATraj-OS	4,285	0.17	0.14	73.5
OSWorld-Human	6,010	0.25	0.21	108.4
AgentNet	16,741	0.71	0.58	278.4
Union	27,765	1.17	0.99	464.5

Table 4: RQ4 profile-construction cost. Times are three-run medians, and resident set size (RSS) is the largest peak observed for the semantic profile.

Both time curves are monotonic across the five input sizes. The semantic curve has a descriptive slope of 0.0422 ms per operation ($R^2 = 0.9997$), reaching 23,731 operations/s on the union. For the 27,765-operation union, semantic construction completes in 1.17 seconds, reaches 464.5 MiB peak RSS, and incurs 180 ms (18.2%) more time and 6.0 MiB (1.3%) more peak RSS than raw-action grouping. As a direct replay of the adopted automatic backend, the fixed 20,866-operation A2 input constructs either its operation-count or 494,862,929-token profile in 0.61 s median, with at most

308.9 MiB observed RSS. Both outputs load in stock pprof and conserve their exact additive mass. This replay measures the profiling core after marks are fixed; automatic annotation latency is separate.

Scope and Limitations

Results apply to the named populations and annotation protocols. CodeTraceBench is the complete development population for the Agent-assisted constructor; OSWorld-Human separately evaluates recurrence and supervised boundary backends. RQ2 scores released localizer predictions over complete public workloads; it does not measure end-user diagnosis time. RQ4 measures fixed-input parsing, stack construction, folding, and serialization, excluding capture, source adaptation, automatic annotation, and live-agent overhead.

Related Work

Agent observability. Platforms aggregate spans, metadata, and evaluations (LangChain 2024; Langfuse 2024; OpenTelemetry 2024; Datadog 2025). Datadog Patterns and LangSmith Insights derive cross-trace hierarchies and roll up metrics, NeMo profiles instrumented workflows, and OpenTelemetry Profiles links profiles to traces (Datadog 2026; LangChain 2026; NVIDIA 2026; OpenTelemetry 2026). These systems establish grouping, rollups, profile naming and compatibility, and trace linkage. AGENTPROF complements these systems by recursively assigning shared semantic responsibility over a preserved source-call tree and folding conserved resource measures into standard pprof profiles.

Cross-run agent analysis. TraceProbe supplies resource-aware process profiles, while Graphectory builds per-trajectory process graphs with aggregated phase patterns (Shu et al. 2026; Liu et al. 2026a). Hodoscope compares behavior distributions to direct human review, while TraceGraph builds shared decision landscapes that guide recovery (Zhong, Saxena, and Raghunathan 2026; Nian et al. 2026). Together, these systems establish canonical actions, recurring cross-run structure, and analysis-to-decision precedents. AGENTPROF complements them with variable-depth responsibility annotations that cover native trace evidence and can replay the same boundaries under different additive resource measures.

Profiling and diagnosis. pprof and flame graphs fold runtime stacks, whereas Pivot Tracing dynamically selects and groups measurements across causally related events (Google 2024; Gregg 2011; Mace, Roelke, and Fonseca 2015). AgentRx diagnoses failures, CodeTracer reconstructs per-run states, and localization benchmarks score faulty steps, agents, or expert-supported perspectives (Barke et al. 2026; Li et al. 2026; Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a; In et al. 2026). AGENTPROF applies the profiler’s population-level principle to agents by annotating recurring semantic responsibility over completed trajectories and folding it without requiring code identity or treating runtime nesting as semantic truth.

Conclusion

Agent observability needs profiling, not only debugging. AGENTPROF recursively annotates semantic responsibility over source-native traces and folds covered source evidence under shared operation stacks across runs. Its profiles expose resource-dependent bottlenecks and success–failure differences, improve stage-partition agreement and localization MAP over raw action, and recover OSWorld-Human groups at 0.680 boundary F1 and 0.786 B³ F1. AGENTPROF builds a 27,765-operation profile in 1.17 seconds, making population profiling practical alongside per-run debugging.

References

- Abhyankar, R.; Qi, Q.; and Zhang, Y. 2025. OSWorld-Human: Benchmarking the Efficiency of Computer-Use Agents. arXiv preprint arXiv:2506.16042.
- Anthropic. 2025. Claude Code: An Agentic Coding Tool. <https://code.claude.com/docs/en/overview>.
- Arize AI. 2024. OpenInference Specification. <https://arize-ai.github.io/openinference/spec/>.
- Bagga, A.; and Baldwin, B. 1998. Entity-Based Cross-Document Coreferencing Using the Vector Space Model. In *36th Annual Meeting of the Association for Computational Linguistics and 17th International Conference on Computational Linguistics, Volume 1*, 79–85. Montreal, Quebec, Canada: Association for Computational Linguistics.
- Barke, S.; Goyal, A.; Khare, A.; Singh, A.; Nath, S.; and Bansal, C. 2026. AgentRx: Diagnosing AI Agent Failures from Execution Trajectories. arXiv preprint arXiv:2602.02475.
- Bouma, G. 2009. Normalized (Pointwise) Mutual Information in Collocation Extraction. In *From Form to Meaning: Processing Texts Automatically, Proceedings of the Biennial GSCS Conference 2009*, 31–40. Tübingen, Germany.
- Bouzenia, I.; and Pradel, M. 2025. Understanding Software Engineering Agents: A Study of Thought-Action-Result Trajectories. In *2025 IEEE/ACM 40th International Conference on Automated Software Engineering (ASE)*, 2846–2857. IEEE.
- Chen, M.; Wang, J.; Mu, F.; Wang, Y.; Liu, Z.; Feng, H.; and Wang, Q. 2026a. Seeing the Whole Elephant: A Benchmark for Failure Attribution in LLM-Based Multi-Agent Systems. arXiv preprint arXiv:2604.22708.
- Chen, X.; Yin, Z.; He, S.; Huang, B.; Lei, S.; et al. 2026b. Safactory: A Scalable Agentic Infrastructure for Training Trustworthy Autonomous Intelligence. arXiv preprint arXiv:2605.06230.
- Datadog. 2025. LLM Observability. https://docs.datadoghq.com/llm_observability/.
- Datadog. 2026. LLM Observability Patterns. https://docs.datadoghq.com/llm_observability/monitoring/patterns/.
- Deng, X.; Gu, Y.; Zheng, B.; Chen, S.; Stevens, S.; Wang, B.; Sun, H.; and Su, Y. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

- Fan, S.; Ye, X.; Huo, Y.; Chen, Z.-Y.; Guo, Y.; Yang, S.; Yang, W.; Ye, S.; Chen, J.; Chen, H.; Cong, X.; and Lin, Y. 2026. AgentProcessBench: Diagnosing Step-Level Process Quality in Tool-Using Agents. In *Proceedings of KDD*.
- Google. 2024. pprof. <https://github.com/google/pprof>.
- Gregg, B. 2011. Flame Graphs. <https://www.brendangregg.com/flamegraphs.html>.
- In, Y.; Tanjim, M.; Subramanian, J.; Kim, S.; Bhattacharya, U.; Kim, W.; Park, S.; Sarkhel, S.; and Park, C. 2026. Rethinking Failure Attribution in Multi-Agent Systems: A Multi-Perspective Benchmark and Evaluation. arXiv:2603.25001.
- LangChain. 2024. LangSmith Observability. <https://docs.langchain.com/langsmith/observability>.
- LangChain. 2026. LangSmith Insights. <https://docs.langchain.com/langsmith/insights>.
- Langfuse. 2024. Langfuse Observability. <https://langfuse.com/docs/observability/overview>.
- Lewis, D. D.; Yang, Y.; Rose, T. G.; and Li, F. 2004. RCV1: A New Benchmark Collection for Text Categorization Research. *Journal of Machine Learning Research*, 5: 361–397.
- Li, H.; Yao, Y.; Zhu, L.; Feng, R.; Ye, H.; Wang, J.; He, Y.; Zou, P.; Zhang, L.; Lei, X.; Huang, H.; Deng, K.; Sun, M.; Zhang, Z.; Ye, H.; and Liu, J. 2026. CodeTracer: Towards Traceable Agent States. arXiv:2604.11641.
- Li, M.; Zhao, Y.; Yu, B.; Song, F.; Li, H.; Yu, H.; Li, Z.; Huang, F.; and Li, Y. 2023. API-Bank: A Comprehensive Benchmark for Tool-Augmented LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 3102–3116.
- Li, W.; Bishop, W. E.; Li, A.; Rawles, C.; Campbell-Ajala, F.; Tyamagundlu, D.; and Riva, O. 2024. On the Effects of Data Scale on UI Control Agents. In *NeurIPS 2024, Datasets and Benchmarks Track*.
- Liu, S.; Chen, Y.; Krishna, R.; Sinha, S.; Ganhotra, J.; and Jabbarvand, R. 2026a. Process-Centric Analysis of Agentic Software Systems. *Proceedings of the ACM on Programming Languages*, 10(OOPSLA1): 1961–1988.
- Liu, Z.; Xie, J.; Ding, Z.; Li, Z.; Yang, B.; et al. 2026b. ScaleCUA: Scaling Open-Source Computer Use Agents with Cross-Platform Data. In *Proceedings of ICLR*.
- Lu, Q.; Shao, W.; Liu, Z.; Meng, F.; Li, B.; Chen, B.; Huang, S.; Zhang, K.; Qiao, Y.; and Luo, P. 2025. GUIOdyssey: A Comprehensive Dataset for Cross-App GUI Navigation on Mobile Devices. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
- Lù, X. H.; Kasner, Z.; and Reddy, S. 2024. WebLINX: Real-World Website Navigation with Multi-Turn Dialogue. In *Proceedings of ICML*, volume 235 of *Proceedings of Machine Learning Research*, 33007–33056. PMLR.
- Lù, X. H.; Kazemnejad, A.; Meade, N.; Patel, A.; Shin, D.; Zambrano, A.; Stańczak, K.; Shaw, P.; Pal, C. J.; and Reddy, S. 2025. AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories. In *Conference on Language Modeling (COLM)*.
- Ma, C.; Zhang, J.; Zhu, Z.; Yang, C.; Yang, Y.; Jin, Y.; Lan, Z.; Kong, L.; and He, J. 2024. AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents. In *Advances in Neural Information Processing Systems*, volume 37, 74325–74362. Curran Associates, Inc.
- Mace, J.; Roelke, R.; and Fonseca, R. 2015. Pivot Tracing: Dynamic Causal Monitoring for Distributed Systems. In *Proceedings of the 25th Symposium on Operating Systems Principles*, 378–393.
- MacQueen, J. B. 1967. Some Methods for Classification and Analysis of Multivariate Observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, 281–297. University of California Press.
- McCallum, A.; and Nigam, K. 1998. A Comparison of Event Models for Naive Bayes Text Classification. In *AAAI-98 Workshop on Learning for Text Categorization*, 41–48. AAAI Press.
- Nian, J.; Chen, K.; Zhang, G.; Cao, Y.; and Jiang, Y. 2026. TraceGraph: Shared Decision Landscapes for Diagnosing and Improving Agent Trajectories. arXiv:2605.31308.
- NVIDIA. 2026. Profiling and Performance Monitoring of NVIDIA NeMo Agent Toolkit Workflows. Official documentation.
- OpenAI. 2025. OpenAI Codex CLI. GitHub repository.
- OpenTelemetry. 2024. OpenTelemetry GenAI Semantic Conventions. <https://github.com/open-telemetry/semantic-conventions-genai>.
- OpenTelemetry. 2026. OpenTelemetry Profiles. Official specification.
- Qin, Y.; Liang, S.; Ye, Y.; Zhu, K.; Yan, L.; Lu, Y.; Lin, Y.; Cong, X.; Tang, X.; Qian, B.; Zhao, S.; Hong, L.; Tian, R.; Xie, R.; Zhou, J.; Gerber, M.; Li, D.; Liu, Z.; and Sun, M. 2024. ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.
- Qwen Team. 2026. Qwen3.6-27B: Flagship-Level Coding in a 27B Dense Model. Official model card.
- Robertson, S. 2008. A New Interpretation of Average Precision. In *Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, 689–690. ACM.
- Rosenberg, A.; and Hirschberg, J. 2007. V-Measure: A Conditional Entropy-Based External Cluster Evaluation Measure. In *Proceedings of the 2007 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, 410–420. Prague, Czech Republic: Association for Computational Linguistics.
- Ruokolainen, T.; Kohonen, O.; Sirts, K.; Grönroos, S.-A.; Kurimo, M.; and Virpioja, S. 2016. A Comparative Study of Minimally Supervised Morphological Segmentation. *Computational Linguistics*, 42(1): 91–120.
- Sajadi, A.; Nguyen, T.; Huynh, K.; Parra, E.; and Chatterjee, P. 2026. TraceView: Interactive Visualization of Agentic Program Repair Trajectories. arXiv:2606.22110.

Shu, R.; Chong, C. Y.; Zhou, X.; Peng, Y.; Wu, Z.; Han, X.; Zhuang, Z.; Yuan, G.; and Wang, Y. 2026. What Resolve Rate Hides: Trajectory Structure Diagnostics for Coding Agents. arXiv:2607.06184.

Wang, J.; Hou, J.; Wang, F.; Jian, P.; Bao, C.; and Lv, Z. 2026. HINTBench: Horizon-Agent Intrinsic Non-Attack Trajectory Benchmark. arXiv preprint arXiv:2604.13954.

Wang, R.; Jansen, P.; Côté, M.-A.; and Ammanabrolu, P. 2022. ScienceWorld: Is Your Agent Smarter than a 5th Grader? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 11279–11298. Association for Computational Linguistics.

Wang, X.; Wang, B.; Lu, D.; Yang, J.; Xie, T.; et al. 2025. AgentNet. <https://huggingface.co/datasets/xlangai/AgentNet>.

Wilson, E. B. 1927. Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158): 209–212.

Xie, T.; Zhang, D.; Chen, J.; Li, X.; Zhao, S.; Cao, R.; Hua, T. J.; Cheng, Z.; Shin, D.; Lei, F.; Liu, Y.; Xu, Y.; Zhou, S.; Savarese, S.; Xiong, C.; Zhong, V.; and Yu, T. 2024. OS-World: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. In *Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track*.

Xu, Y.; Lu, D.; Shen, Z.; Wang, J.; Wang, Z.; Mao, Y.; Xiong, C.; and Yu, T. 2025. AgentTrek: Agent Trajectory Synthesis via Guiding Replay with Web Tutorials. In *Proceedings of ICLR*.

Yang, J.; Jimenez, C. E.; Wettig, A.; Lieret, K.; Yao, S.; Narasimhan, K.; and Press, O. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Yao, S.; Chen, H.; Yang, J.; and Narasimhan, K. 2022. Web-Shop: Towards Scalable Real-World Web Interaction with Grounded Language Agents. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*.

Yao, S.; Shinn, N.; Razavi, P.; and Narasimhan, K. 2025. τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *The Thirteenth International Conference on Learning Representations (ICLR)*.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

Zheng, Y.; Hu, Y.; Yu, T.; and Quinn, A. 2025. AgentSight: System-Level Observability for AI Agents Using eBPF. In *Proceedings of the 4th Workshop on Practical Adoption Challenges of ML for Systems*.

Zhong, Z.; Saxena, S.; and Raghunathan, A. 2026. Hodoscope: Unsupervised Monitoring for AI Misbehaviors. arXiv:2604.11072.